

Day 08 – Cloud Server Setup:Nginx & Web Deployment

Part 1 – Launch Cloud Instance & SSH Access

Step 1: Login to AWS

Step 2: Launch EC2 Instance

Click:

[Launch Instance](#)

Configure:

Setting	Value
Name	day08-nginx-server
AMI	Ubuntu Server 22.04 LTS
Instance Type	t2.micro (Free Tier)
Key Pair	Create new key pair
Security Group	Allow SSH + HTTP

Step 3: Create Key Pair

Choose:

- Key Pair Name: `day08-key`
- Type: `.pem`

Download the `.pem` file carefully.

Example:

`day08-key.pem`

Store it safely in Downloads folder.

Step 4: Configure Security Group

Enable these:

Type	Port
------	------

SSH	22
-----	----

HTTP	80
------	----

This allows:

- SSH access
- Website access from browser

Click:

[Launch Instance](#)

Step 5: Copy Public IP

After instance launches:

Copy:

Public IPv4 address

Example:

❏ 13.233.xx.xx

❏

Step 6: Connect via SSH

Windows (Git Bash / PowerShell)

Move to Downloads folder:

❏ `cd ~/Downloads`

❏ Give key permission:

❏ `chmod 400 day08-key.pem`

❏ Connect to server:

❏ `ssh -i day08-key.pem ubuntu@YOUR_PUBLIC_IP`

❏ Example:

❏ `ssh -i day08-key.pem ubuntu@13.233.xx.xx`

❏

Expected Output

You should see something like:

❏ Welcome to Ubuntu 22.04 LTS
ubuntu@ip-172-31-x-x:~\$

🖼️ 📷 Take Screenshot:

- SSH terminal connected to EC2

```
ubuntu@ip-172-31-39-234: ~  
ubuntu@ip-172-31-39-234:~$ exit  
logout  
Connection to ec2-13-60-55-147.eu-north-1.compute.amazonaws.com closed.  
  
khatu@Parthavi_Khatu MINGW64 ~/Downloads  
$ ssh -i "Key-pair-a.pem" ubuntu@ec2-13-60-55-147.eu-north-1.compute.amazonaws.com  
  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat May 23 13:27:57 UTC 2026  
  
System load:  0.08           Temperature:   -273.1 C  
Usage of /:   30.4% of 6.61GB Processes:      120  
Memory usage: 27%           Users logged in: 0  
Swap usage:   0%            IPv4 address for ens5: 172.31.39.234  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Sat May 23 13:27:58 2026 from 115.96.218.82  
ubuntu@ip-172-31-39-234:~$  
ubuntu@ip-172-31-39-234:~$
```

Part 2 – Install Docker & Nginx

Step 1: Update System

Run:

📄 `sudo apt update && sudo apt upgrade -y`



Step 2: Install Docker

Install Docker:

☐ `sudo apt install docker.io -y`

☐ Start Docker:

☐ `sudo systemctl start docker`

☐ Enable Docker on boot:

☐ `sudo systemctl enable docker`

☐ Check status:

☐ `sudo systemctl status docker`

☐ Expected:

☐ `active (running)`



Step 3: Install Nginx

Install Nginx:

☐ `sudo apt install nginx -y`

☐ Start Nginx:

☐ `sudo systemctl start nginx`

☐ Enable Nginx:

☐ `sudo systemctl enable nginx`

☐ Check status:

❑ `sudo systemctl status nginx`

❑ Expected:

❑ `active (running)`

❑

```
ubuntu@ip-172-31-39-234:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-05-23 13:32:30 UTC; 36s ago
     Invocation: 5825946b144d47aaa5498a0243eda69c
       Docs: man:nginx(8)
    Main PID: 12648 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 41ms
    CGroup: /system.slice/nginx.service
            └─12648 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─12651 "nginx: worker process"
                └─12652 "nginx: worker process"

May 23 13:32:30 ip-172-31-39-234 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 23 13:32:30 ip-172-31-39-234 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-39-234:~$
```

Part 3 – Verify Web Access

Open browser:

❑ `http://YOUR_PUBLIC_IP`

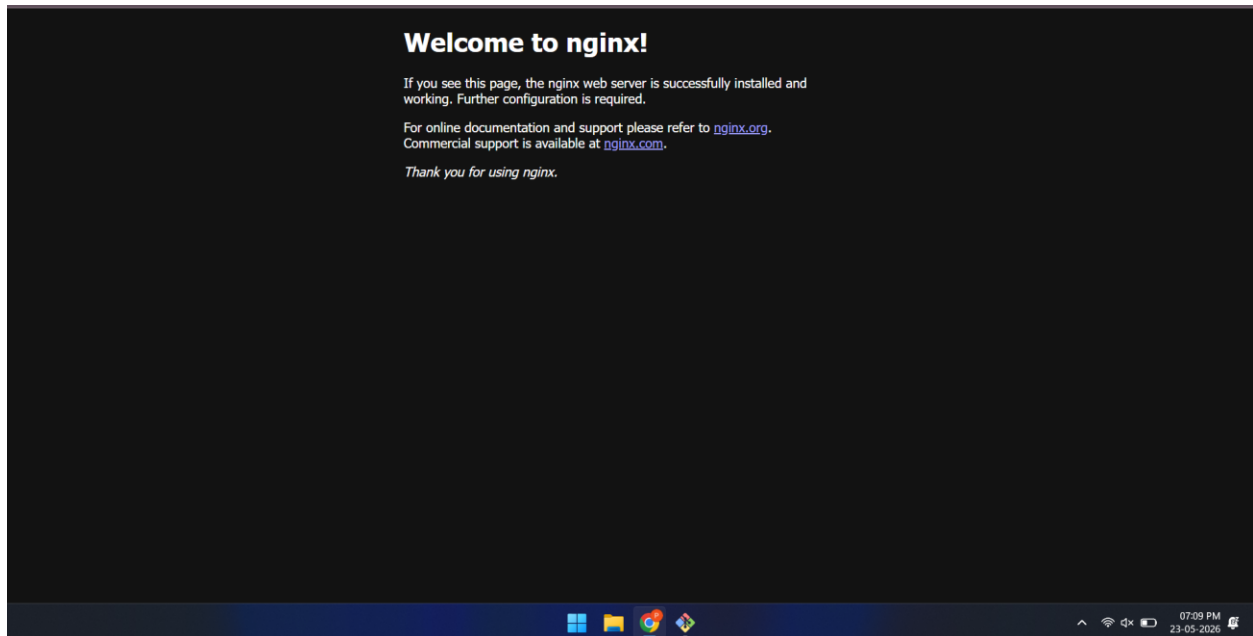
❑ Example:

❑ `http://13.233.xx.xx`

❑

Expected Result

You should see:



“Welcome to nginx!”

 Take Screenshot:

- Nginx welcome page in browser

Part 4 – Extract Nginx Logs

Step 1: View Nginx Logs

Access log:

```
❏ sudo cat /var/log/nginx/access.log
```

❏

```
ubuntu@ip-172-31-39-234:~$ sudo cat /var/log/nginx/access.log
27.102.121.17 - - [23/May/2026:13:38:09 +0000] "GET / HTTP/1.0" 200 615 "-" "ivre-masscan/1.3 https://github.com/robertdavidgraham/"
:::1 - - [23/May/2026:13:38:11 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.18.0"
115.96.218.82 - - [23/May/2026:13:39:05 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/148.0.0
115.96.218.82 - - [23/May/2026:13:39:09 +0000] "GET /favicon.ico HTTP/1.1" 404 197 "http://13.60.55.147/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHT
i/537.36"
ubuntu@ip-172-31-39-234:~$ |
```

Error log:

❑ `sudo cat /var/log/nginx/error.log`

❑ You will see browser request entries.

Step 2: Save Logs to File

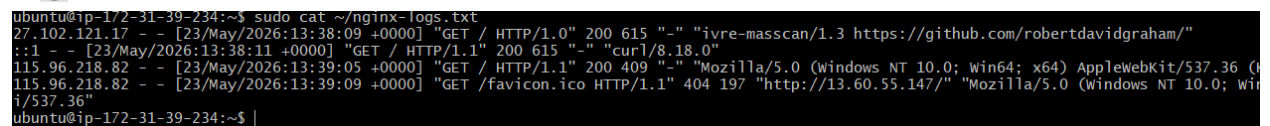
Copy access logs:

❑ `sudo cp /var/log/nginx/access.log ~/nginx-logs.txt`

❑ Verify:

❑ `cat ~/nginx-logs.txt`

❑ 📸 Take Screenshot:

A terminal window screenshot showing the output of the command 'cat ~/nginx-logs.txt'. The output displays several log entries from the nginx access log, including IP addresses, timestamps, HTTP methods, status codes, and user agents. The entries are separated by newlines and some are truncated on the right side.

```
ubuntu@ip-172-31-39-234:~$ sudo cat ~/nginx-logs.txt
27.102.121.17 - - [23/May/2026:13:38:09 +0000] "GET / HTTP/1.0" 200 615 "-" "ivre-masscan/1.3 https://github.com/robertdavidgraham/"
::1 - - [23/May/2026:13:38:11 +0000] "GET / HTTP/1.1" 200 615 "-" "curl/8.18.0"
115.96.218.82 - - [23/May/2026:13:39:05 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/115.0.0.0 Safari/537.36"
115.96.218.82 - - [23/May/2026:13:39:09 +0000] "GET /favicon.ico HTTP/1.1" 404 197 "http://13.60.55.147/" "Mozilla/5.0 (Windows NT 10.0; Win
i/537.36"
ubuntu@ip-172-31-39-234:~$
```

- Terminal showing nginx-logs.txt contents
-

Step 3: Download Log File to Local Machine

Open NEW terminal on your local machine.

Move to Downloads:

❑ `cd ~/Downloads`

❑ Run:

❑ `scp -i day08-key.pem ubuntu@YOUR_PUBLIC_IP:~/nginx-logs.txt .`

❑ Example:

❑ `scp -i day08-key.pem ubuntu@13.233.xx.xx:~/nginx-logs.txt .`

❑ Check downloaded file:

❑ `ls`

❑ You should see:

❑ `nginx-logs.txt`

❑

Nginx-logs.txt-----

ubuntu@ip-172-31-39-234:~\$ sudo cat nginx-logs.txt

27.102.121.17 - - [23/May/2026:13:38:09 +0000] "GET / HTTP/1.0" 200

615 "-" "ivre-masscan/1.3 https://github.com/robertdavidgraham/"

:::1 - - [23/May/2026:13:38:11 +0000] "GET / HTTP/1.1" 200 615 "-"

"curl/8.18.0"

115.96.218.82 - - [23/May/2026:13:39:05 +0000] "GET / HTTP/1.1" 200

409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36

(KHTML, like Gecko) Chrome/148.0.0.0 Safari/537.36"

115.96.218.82 - - [23/May/2026:13:39:09 +0000] "GET /favicon.ico

HTTP/1.1" 404 197 "http://13.60.55.147/" "Mozilla/5.0 (Windows NT

10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)

Chrome/148.0.0.0 Safari/537.36"